

DESIGN PORTFOLIO

Aryaman Bhatnagar

CAD Designs · CFD & FEA Simulations

PORTFOLIO SUMMARY

F1 2026 Livery	SOLIDWORKS + CFD simulation
F1 2022 to 2025 Livery	SOLIDWORKS · Ground effect
F1 2018 to 2021 Livery	PTC Creo · First F1 project
Electric Motorcycle	SOLIDWORKS · Featured on DS
Sports Motorcycle	Siemens NX · Sub-D surfacing
Motorcycle CFD	Ansys Fluent · Thesis CFD study
Sedan FEA + CFD	Siemens NX + Ansys Static + Fluent
UAV Design	Siemens NX + Ansys Static

SOLIDWORKS and CATIA

Siemens NX

PTC Creo

Fusion 360

Ansys Fluent

Ansys Mechanical

OPENFOAM

ALTAIR Hypermesh

SKILLS & SOFTWARE EXPERTISE

SOLIDWORKS & CATIA

CAD / FEA

Part Design · Weldments · Sheet Metal · Surfacing
· Generative Shape Design (GSD)

Siemens NX

Advanced CAD

Pipe Routing · Freeform Surface Modelling · Solid
Modelling · Assembly · Additive Manufacturing

PTC Creo and Fusion 360

Generative CAD

Freestyle / Advanced Surfacing · T-Spline Modelling ·
Rendering · CAM

Ansys Fluent

CFD

Turbulence Modelling · Contours · Postprocessing
(Pressure, Velocity, Turbulence, Vorticity) · Mesh
Independence Study

Ansys Mechanical, HYPERMESH & 3DX

FEA

Static Structural · Deformation · Stress Analysis
Explicit Dynamics · Topology Optimisation · Modal
Analysis

Engineering Drawing

Manufacturing Drawing

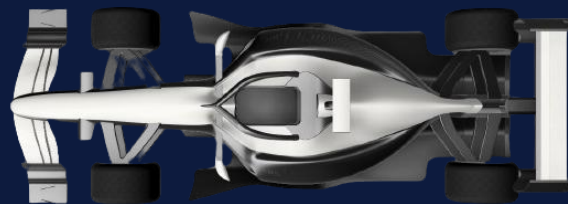
Assembly Drawing

Technical Drawings · Tolerance Standards · GD&T

01

FORMULA-1 CARS

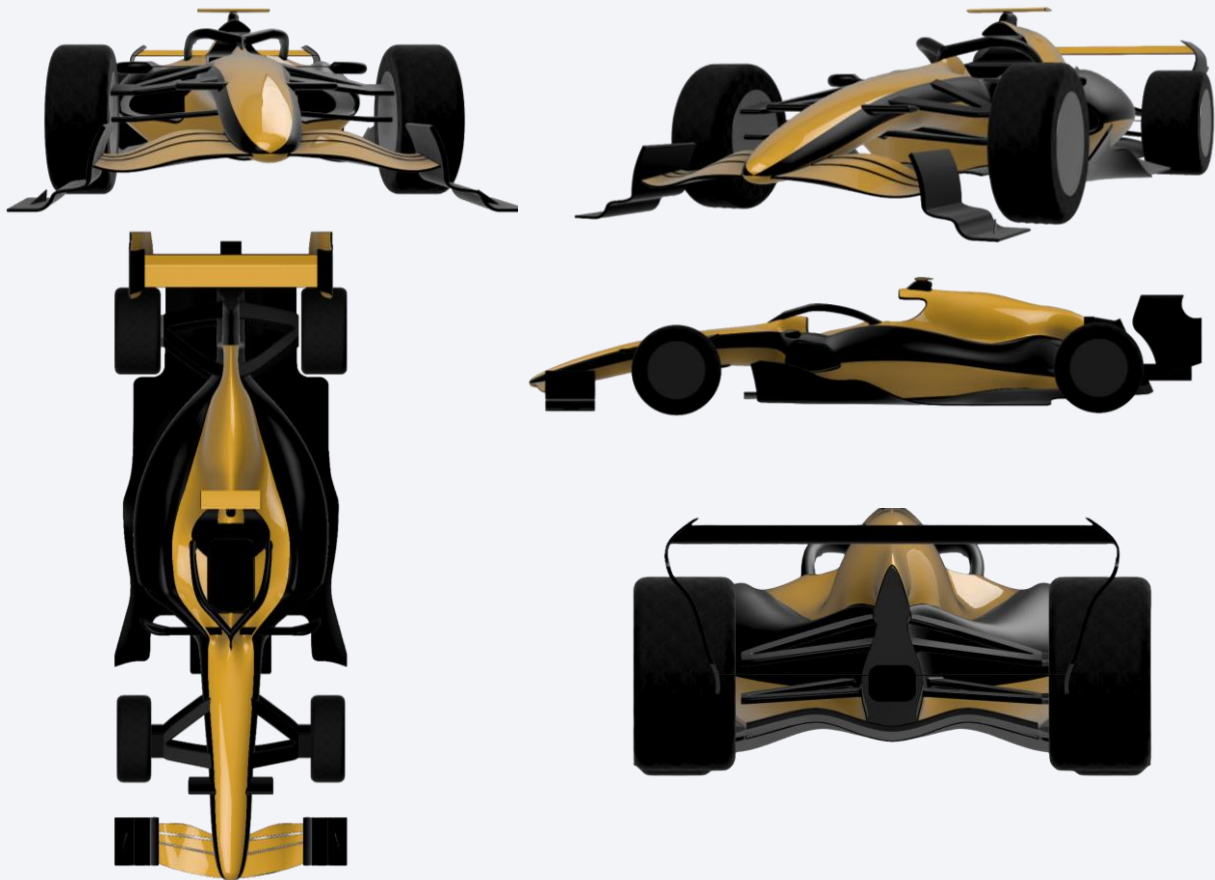
2018 · 2022 · 2026 FIA Technical Regulations



FORMULA-1 CAR 2026 REGULATION

PROJECT OVERVIEW

A concept Formula 1 car developed based on 2026 FIA Technical Regulations, focusing on aerodynamic surface design and flow structure analysis using CFD.



KEY FEATURES & HIGHLIGHTS

- 1 Parametric surfacing applied to monocoque, sidepods, and floor geometry
- 2 Structural elements modelled using simplified profiles for conceptual representation
- 3 Aero surfaces: multi-element front wing, active rear wing concept
- 4 Followed by full CFD simulation in Ansys Fluent 2024 R2

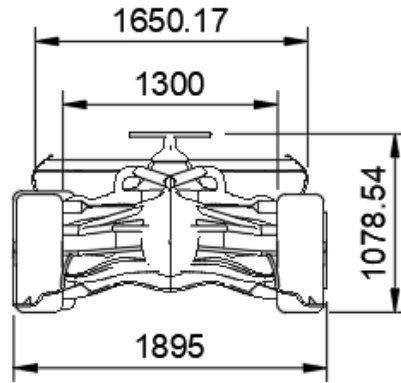
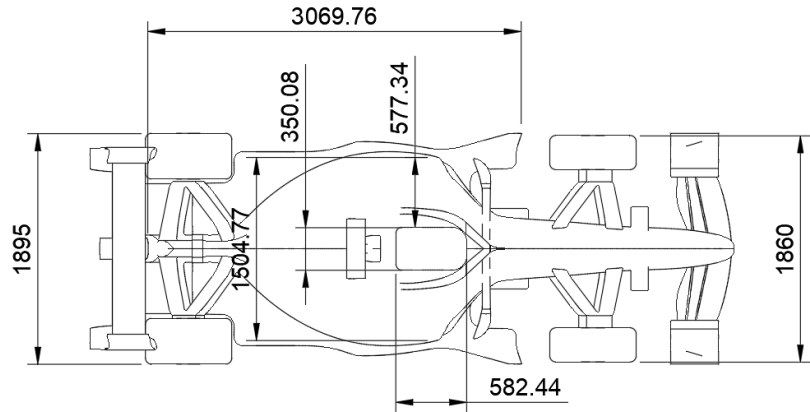
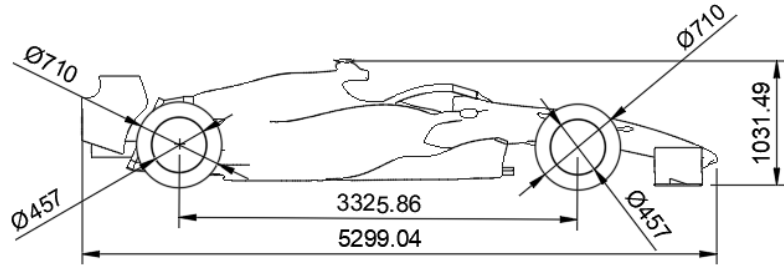
TECHNICAL SPECIFICATIONS

Regulation Year	2026 FIA Technical Reg.
CAD Software	SOLIDWORKS 2024
Chassis Method	Surfacing + Weldments
Drawings	GD&T (FIA spec)
Aero Package	Front wing, Floor, Rear wing

FORMULA-1 CAR 2026 ENGINEERING DRAWING

PROJECT OVERVIEW

A concept Formula 1 car developed based on 2026 FIA Technical Regulations, focusing on aerodynamic surface design and flow structure analysis using CFD.



KEY FEATURES & HIGHLIGHTS

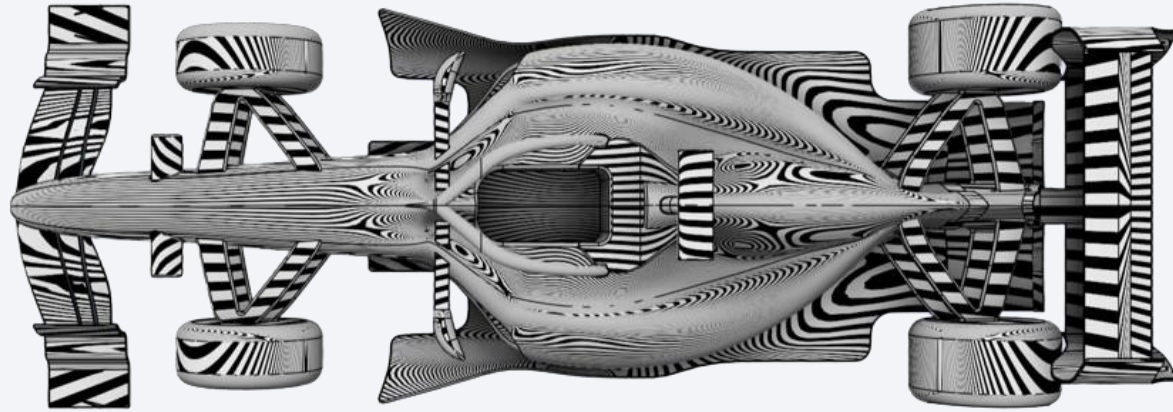
- 1 F1 concept design follows a compliant overall width of 1,895 mm and wheelbase package for improved agility and aerodynamics.
- 2 Designed realistic monocoque, cockpit, sidepod, and floor dimensions in compliance with current packaging in F1 and modern safety cell layout.
- 3 Incorporated aerodynamic approach of 2026 F1, with narrow body dimensions, reduced floor dimensions, and compact diffuser design.
- 4 Created a technically sound dimensional analysis from above in the form of top view using motorsport vehicle packaging considerations.

Parameter	FIA Regulation (mm)	My Design (mm)
Overall Width	1,900	1,895
Wheelbase	3,400	3,325.9
Overall Length	No fixed limit	5,299
Tyre Diameter	710	710
Rim Diameter	457	457
Front Wing Width	1,650	1,650.2

FORMULA-1 CAR 2026 ZEBRA ANALYSIS

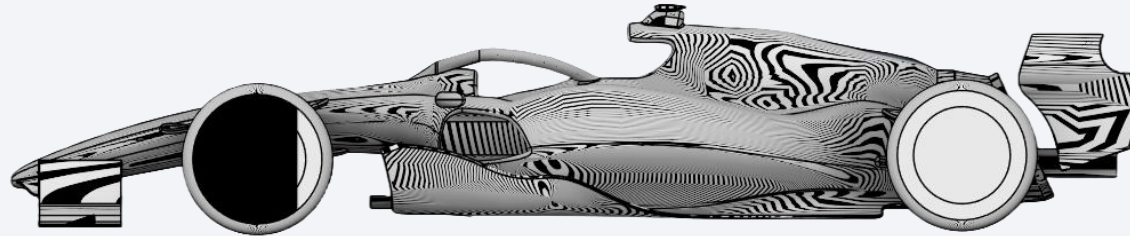
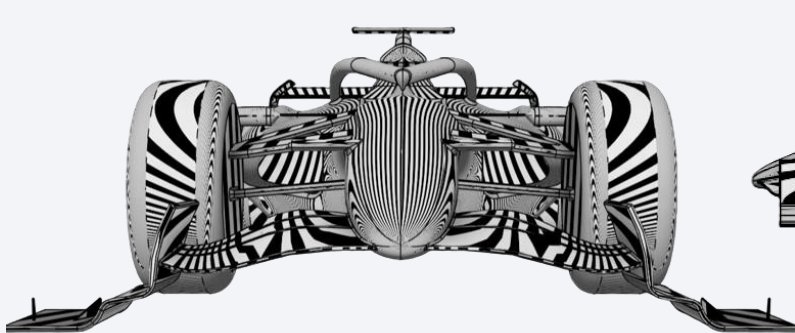
PROJECT OVERVIEW

A concept Formula 1 car developed based on 2026 FIA Technical Regulations, focusing on aerodynamic surface design and flow structure analysis using CFD.



KEY FINDINGS

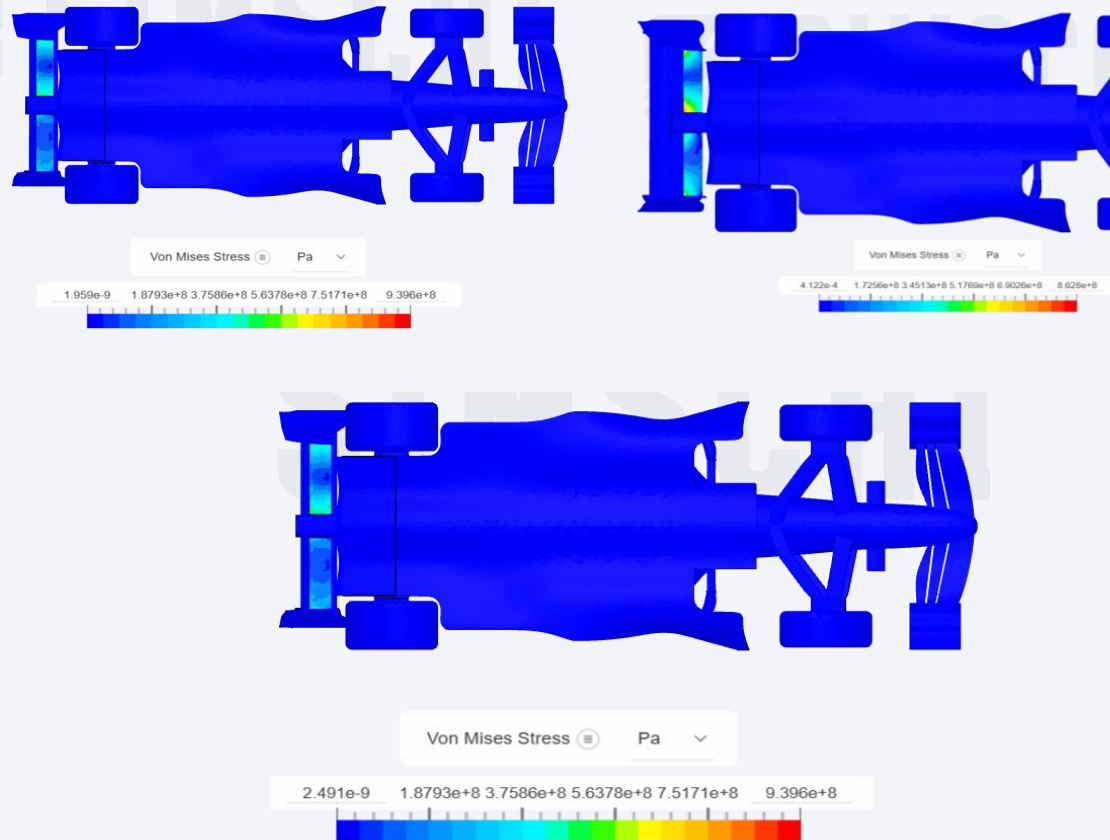
- 1 Curvature Continuity G2 maintained in Monocoque, Sidepod and Engine Cover Floor Geometry
- 2 Zebra Striping Validation proves continuity of curvature between major aerodynamic surfaces
- 3 Surface discontinuities restricted to structural components and sharp-edge aero elements, consistent with FIA regulation geometry constraints.
- 4 Iterative Surface optimisation performed from Concept Design stage onwards



FORMULA-1 CAR 2026 STATIC ANALYSIS

PROJECT OVERVIEW

Static structure analysis of the proposed concept of the chassis for F1 in 2026 analyzed in SimScale with use of F1 grade CFRP (with $E = 1.38 \times 10^{11}$ Pa and Tensile Strength 1200 MPa), subjected to aerodynamic downforces as pressure loads on front wings (2,674 Pa), floor (641 Pa), and rear wings (1,602 Pa) while moving at the speed of 220 km/h race evaluated across three load cases of braking, pothole or bump, and cornering.



KEY FINDINGS

1

Braking scenario: 5,000 N applied along X direction at rear suspension attachment points with front attachment points being constrained; Von Mises stress value of 939.6 MPa recorded at the front suspension area, having a FOS of 1.28

2

Pothole load scenario: 7,500 N applied along Z direction at both left front and rear suspension attachment points with right side being constrained, modelling left side kerb impact; Von Mises stress of 862.8 MPa was observed with asymmetrical torsional stress observed at the left side having a FOS of 1.39

3

Cornering load scenario: lateral loads applied to all suspension attachment points with front attachment points being constrained; Von Mises stress of 862.8 MPa was recorded at the rear wing end plate connection, having a FOS of 1.39

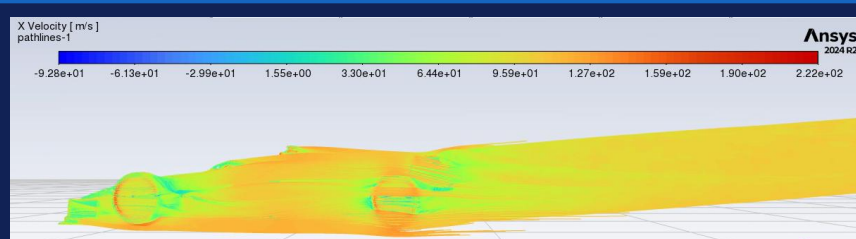
CFD SIMULATION 2026 F1 CAR using ANSYS FLUENT

PROJECT OVERVIEW

Steady-state RANS simulation using k- ω SST model. Velocity inlet at 61 m/s, pressure outlet at 0 Pa, moving ground with rotating wheels. Near-wall treatment with $y^+ \approx 30$

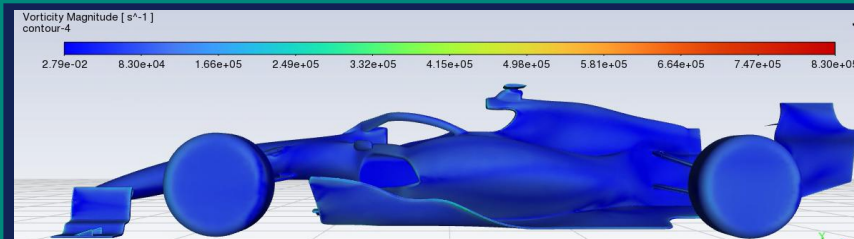
- Model: Steady-state RANS
- Turbulence: k- ω SST
- Freestream velocity: 61.1 m/s (220 km/h)
- Domain: External aerodynamic flow
- Ground condition: (moving floor at 220km/h)

X-Velocity Wake Profile



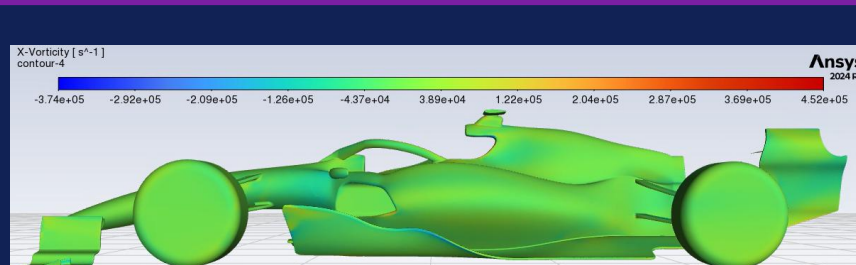
The plot identifies the low-energy region downstream of the rear diffuser. Accelerated underfloor flow confirms ground effect suction consistent with 2026 venturi floor geometry. The wide lateral wake indicates front wing outwash channeling flow around the wheels

Vorticity Magnitude



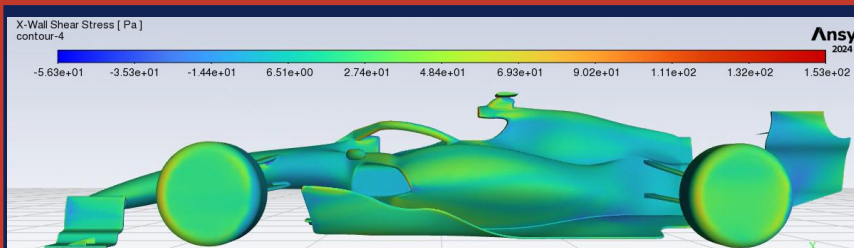
High vorticity concentrated at front wing endplates, floor edges, and rear wing tips confirms vortex generation at all critical aerodynamic surfaces. Low vorticity over primary body surfaces indicates attached flow.

X-Vorticity Distribution



The presence of the X-vorticity in the form of areas where the values are both positive and negative (shown in different red and blue gradient colours) in proximity to the edges of the front wing and the floor denotes counter-rotating vortices, the fundamental mechanism by which F1 floors generate downforce through vortex sealing of the underfloor low-pressure region

X-Wall Shear Stress [Pa]

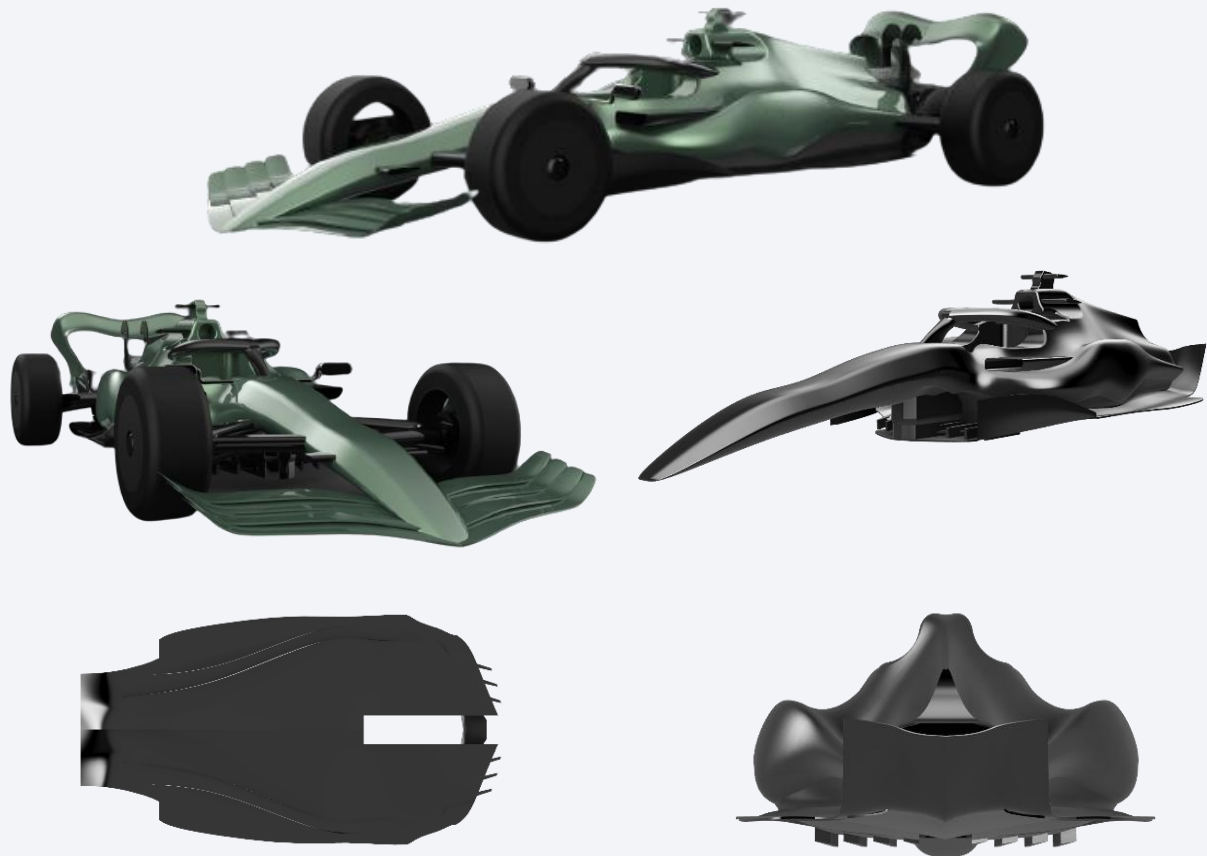


Wall shear stresses characterise the boundary layer adhesion for all aerodynamic surfaces. The large wall shear stresses on the nose, front wing, and floor demonstrate the high energy state of adhesion flow as it moves through the ground effect region. Reduced shear at the rear bodywork is consistent with expected bluff body wake development and flow deceleration.

FORMULA-1 CAR 2022 to 2025 REGULATION

PROJECT OVERVIEW

Ground-up F1 design based on the 2022 FIA Technical Regulations, the era that introduced ground-effect aerodynamics back to the sport.



KEY FEATURES & HIGHLIGHTS

- 1 Ground-effect floor geometry with venturi tunnels and diffuser sections accurately surfaced
- 2 High-sidepod architecture following 2022 regulation aero philosophy
- 3 Distinct multi-element front wing with five-plane cascade design
- 4 GD&T drawings fully annotated to FIA 2022 tolerance specification
- 5 Rendered in British Racing Green with carbon-fibre texture details

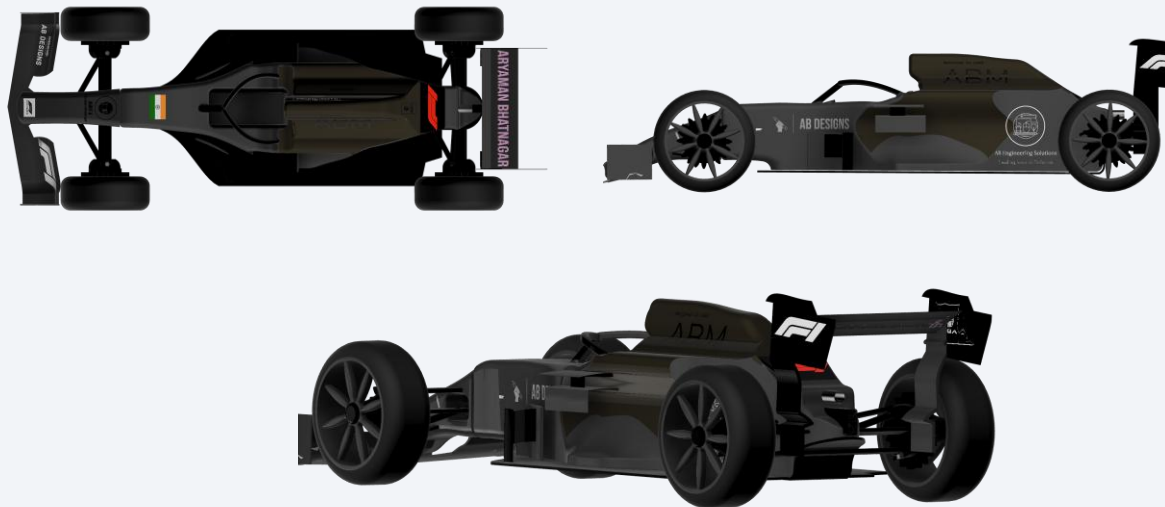
TECHNICAL SPECIFICATIONS

Regulation Year	2022 FIA Technical Reg.
CAD Software	SOLIDWORKS 2024
Aero Era	Ground-effect floor generation
Drawings	GD&T annotated
Key Feature	Multi-element front & rear wings

FORMULA-1 CAR 2018 to 2021 REGULATION

PROJECT OVERVIEW

The earliest of three F1 design iterations, modelled to 2018 FIA Technical Regulations. This project pioneered the author's F1 CAD workflow, establishing a repeatable methodology later refined across 2022 and 2026 designs.

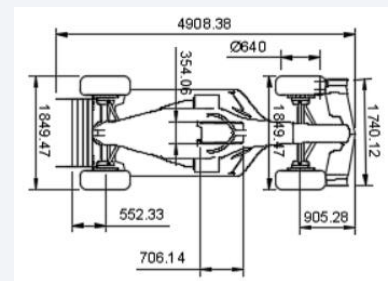
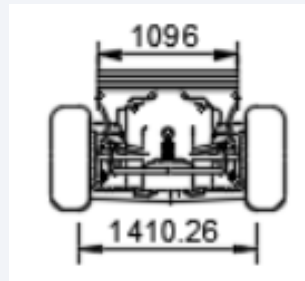
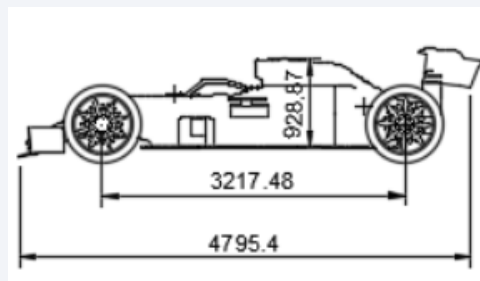


KEY FEATURES & HIGHLIGHTS

- 1 Aerodynamic surfaces created using Style (ISDX) and Boundary Blend tools, ensuring smooth curvature transitions and surface continuity
- 2 Structural components represented using simplified parametric geometry to define suspension layout and overall packaging
- 3 Fully dimensioned 2D drawings including wheelbase 4795 mm, track 1849 mm
- 4 Assembly constructed using top-down design methodology with datum references and constrained component placement

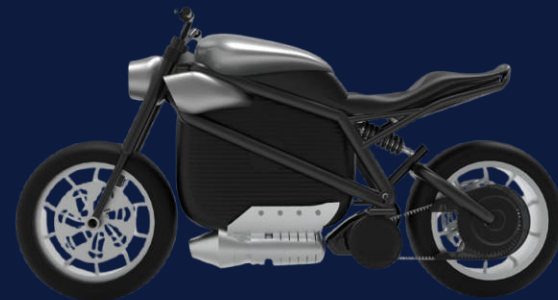
TECHNICAL SPECIFICATIONS

Regulation Year	2018 FIA Technical Reg.
CAD Software	PTC Creo
Chassis	Parametric Design
Aero Surfaces	ISDX and Boundary Blend tool
Drawings	Full dimension + GD&T



02

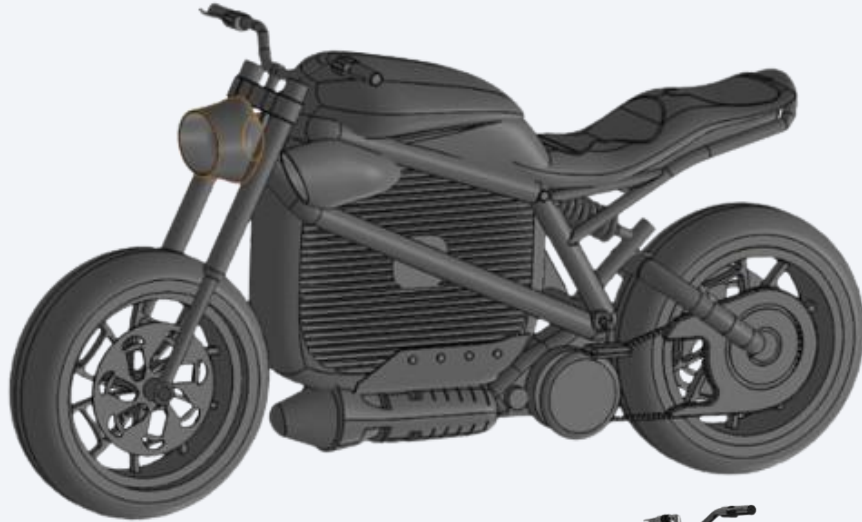
MOTORCYCLE DESIGN



ELECTRIC MOTORCYCLE

PROJECT OVERVIEW

A full-scale electric motorcycle designed from the ground up in SOLIDWORKS. The design was recognised and featured on the official Dassault Systèmes LinkedIn and Instagram pages, highlighting its technical depth and visual quality.



KEY FEATURES & HIGHLIGHTS

- 1 Frame developed using SOLIDWORKS Weldments with parametric structural profiles and trimmed joint intersections
- 2 Sheet metal body panels designed with defined K-factor and bend allowance for manufacturability
- 3 Full assembly comprising 50+ components with constrained mating strategy
- 4 Surface modelling applied to storage tank, seat, and wheel geometry
- 5 Photorealistic rendering using KeyShot for design visualisation

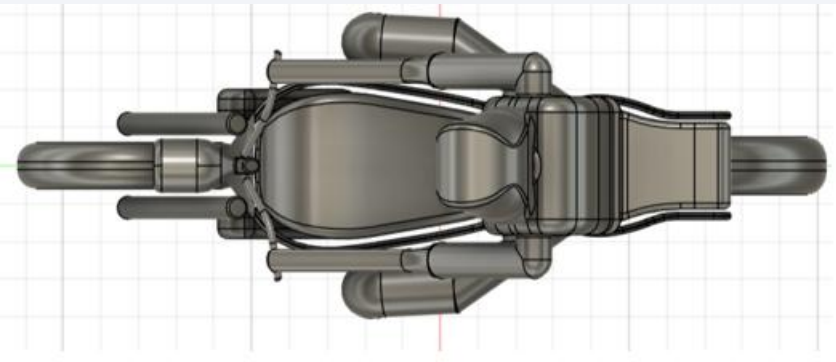
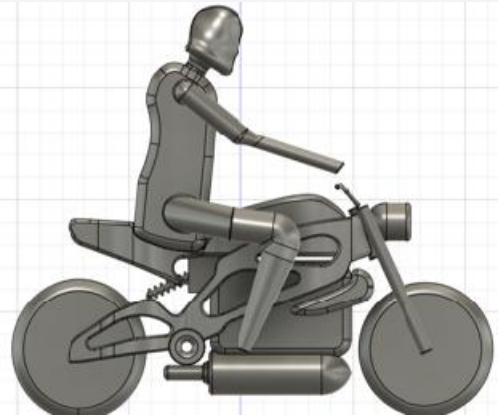
TECHNICAL SPECIFICATIONS

CAD Platform	SOLIDWORKS 2024
Frame Method	Weldments (structural profiles)
Body Method	Sheet Metal (bend-controlled geometry)
Components	Part Design + Assembly
Rendering	KeyShot
Recognition	Dassault Systèmes featured

PROJECT OVERVIEW

DISSERTATION: NAKED TYPE MOTORCYCLE

A detailed motorcycle geometry model developed in SOLIDWORKS, focused on creating an aerodynamically representative naked-type motorcycle suitable for downstream simulation and design evaluation. The project emphasises parametric modelling, surface control, and assembly integration.



KEY FEATURES & HIGHLIGHTS

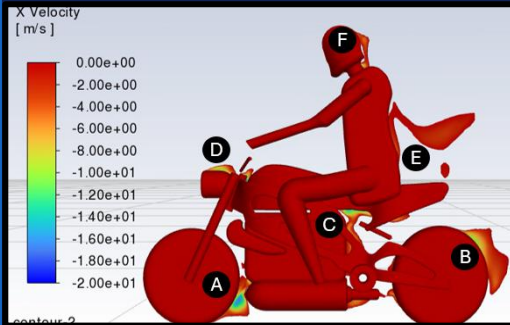
- 1 Parametric CAD modelling approach used to define all major components and ensure design flexibility
- 2 Frame, wheels, and mechanical components created using Extrude, Revolve, and Sweep features
- 3 Fuel tank, seat, and aerodynamic body regions developed using Lofted Surfaces, Boundary Surfaces, and Fillets to ensure smooth transitions
- 4 Full motorcycle assembly created with multiple mated components

TECHNICAL SPECIFICATIONS

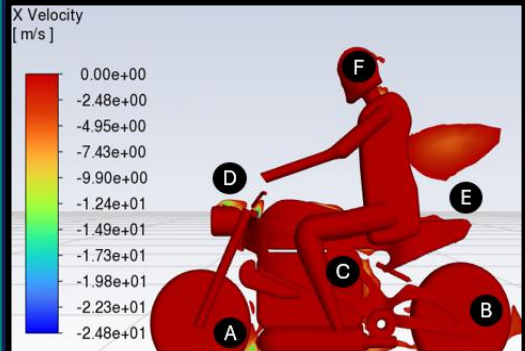
CAD Platform	SOLIDWORKS
Frame Method	Parametric CAD
Body Method	Lofted and Boundary surfaces
Assembly	Full mated components
Meshing	Body of Influence

MSc Dissertation CFD Results

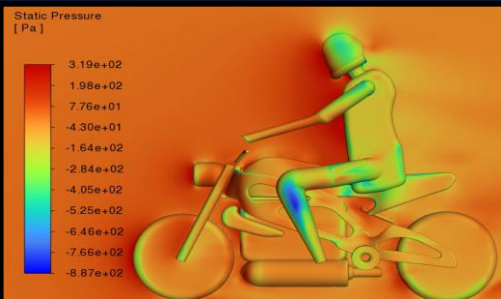
X-Velocity Contour (Baseline)



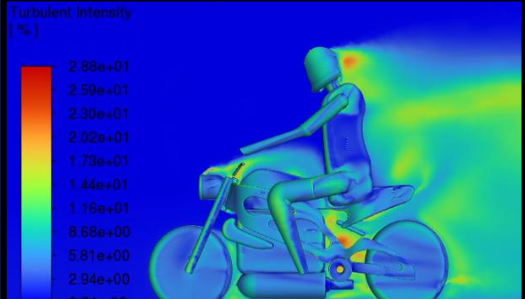
X-Velocity Contour (With VGs)



Static Pressure [Pa]



Turbulent Intensity (%)



Using CFD To Investigate The Effect Of Triangular Vortex Generators On The Aerodynamic Performance Of A Naked-type Motorcycle

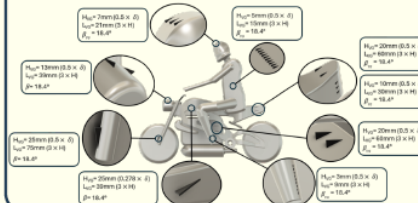
Introduction and Research Gap

As the automotive industry pivots towards sustainability, aerodynamics are crucial for fuel efficiency and emission reduction, primarily by minimising drag. This research aims to fill this critical gap, focusing on improving motorcycle stability and performance through advanced aerodynamic optimisation.

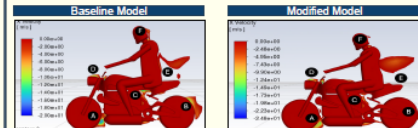
Key Challenges and Research Gap

- Extensive studies on four-wheelers, but motorcycles remain underexplored.
- Motorcycle geometry and rider exposure create unique aerodynamic challenges.

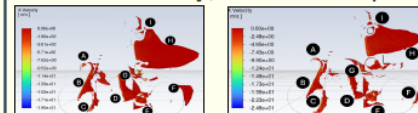
Vortex Generators Placement



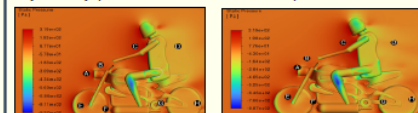
Results and Flow Visualisation



Point A and B, the recirculation bubble is reduced compared to the baseline due to VGs on the front fork re-emerging the boundary layer. Point C, VGs near the battery module reduce backflow, leaving only a thin recirculation zone. Similar effect at Point F, helmet. Point E the VGs on rider's back make the flow stronger, but uniform which reduces unsteady wake

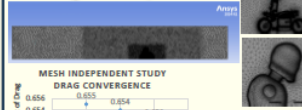


Point A, separation persists but is recirculated downstream, improving uniformity. The VG at the front suspension reduces separation, while a minor improvement is observed in the tank and frame. At point D, the rider's leg shows a more uniform flow due to VG integration. VGs on the rider's head improve flow from the rear of the head to the neck and back, while side body VGs modify upstream airflow and reduce concentrated separation zones.



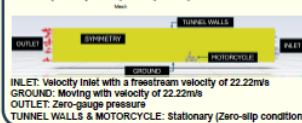
High-pressure stagnation occurs at the point (A), with smoother pressure drops at the top (B) than the baseline. Similar effect at the point (E) and (G), while wakes behind them (F, H) form stronger low-pressure region. Overall, VGs reduce early separation, wakes, and smooth pressure distribution by keeping flow attached and reducing adverse pressure gradients.

Methodology

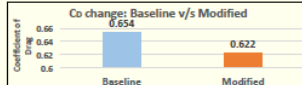


MESH INDEPENDENT STUDY
DRAG CONVERGENCE

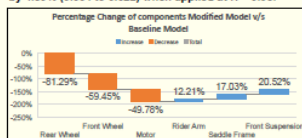
Mesh-3 (0.654), Mesh-4 (0.652) and Mesh-5 (0.645) showed a deviation in C_D values <0.1%, which positively shows convergence between the meshes.



Conclusion



The triangular vortex generator reduced the overall C_D by 4.89% (0.654 to 0.622) when applied at $H = 0.55$.



Components such as the rear wheel registered the highest reduction (-81.29%), followed by the front suspension (-59.45%) and motor (-45.78%). These components were susceptible to flow separation and, therefore, significant improvement can be observed in the components and showing that VGs have effectively delayed boundary layer separation.

Whereas some components, such as the front suspension (20.52%), saddle frame (17.03%) and rider arm (12.21%), have shown an increase in total drag. It can indicate that a modification in the airflow caused by the VGs has led to increased aerodynamic drag in these components.

SPORTS MOTORCYCLE

PROJECT OVERVIEW

A high-performance sports motorcycle modelled in Siemens NX, focusing on advanced surface modelling and geometric control of complex aerodynamic forms.



KEY FEATURES & HIGHLIGHTS

- 1 Sub-D (3D Form) modelling used to generate smooth, continuous aerodynamic fairing surfaces
- 2 Pipe routing tools applied to define frame geometry with controlled bend radii
- 3 Detailed component modelling including drivetrain, suspension, braking system, and wheels
- 4 Rendered in multiple angles with metallic paint & procedural textures

TECHNICAL SPECIFICATIONS

CAD Platform	Siemens NX
Frame Method	Pipe Routing (bend radius control)
Body Method	3D Form Sub-D surface modelling
Surface Quality	High-quality Sub-D surfaces with curvature continuity
Assembly	Full system integration
Rendering	NX Render + post-processing

03

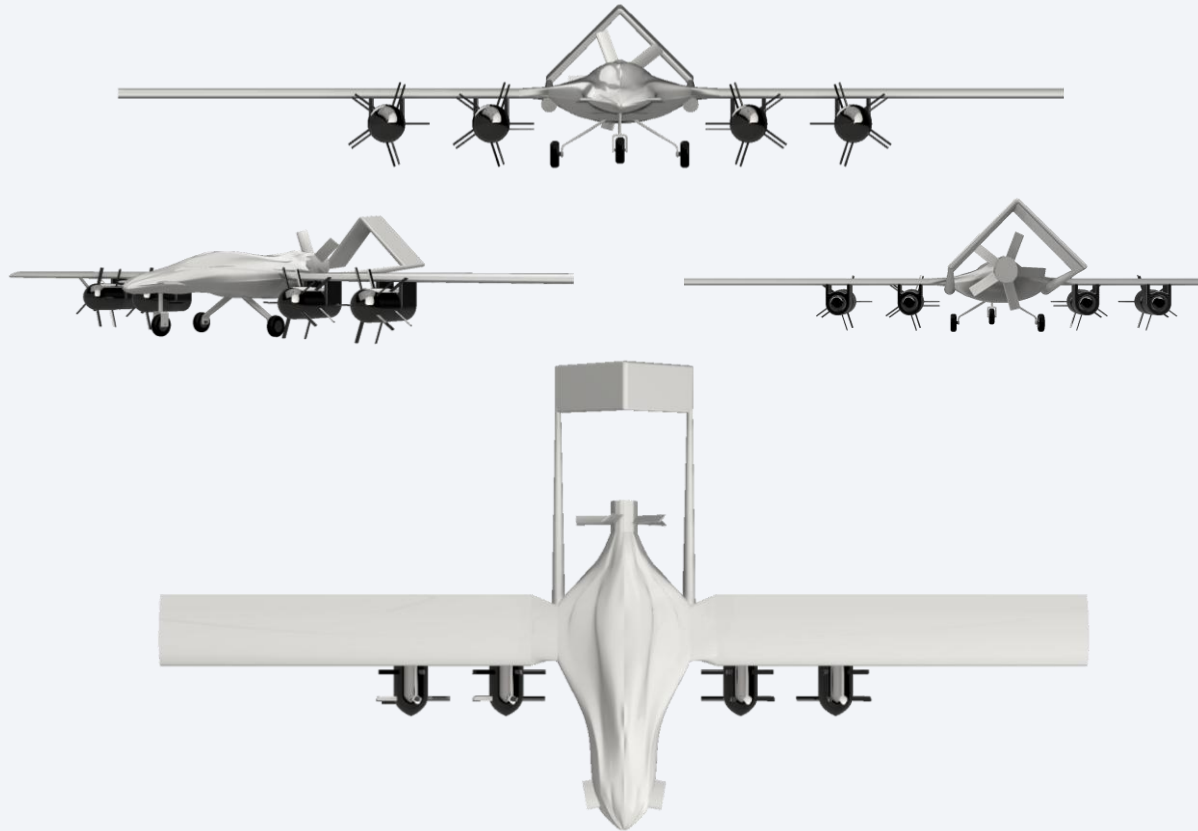
UAV

Surface Modelling · Static Analysis

UAV Design

PROJECT OVERVIEW

Conceptual design and CFD analysis of a fixed-wing pusher-prop UAV, developed in Siemens NX using parametric and Sub-D surface modelling techniques.



KEY FEATURES & HIGHLIGHTS

- 1 Blended wing-fuselage body with rear pusher-prop twin-boom layout for reduced drag and clean payload integration.
- 2 Sub-D surface modelling in Siemens NX achieving G1 continuity, validated through zebra stripe analysis.
- 3 Fully parametric assembly enabling rapid iteration across all airframe components.

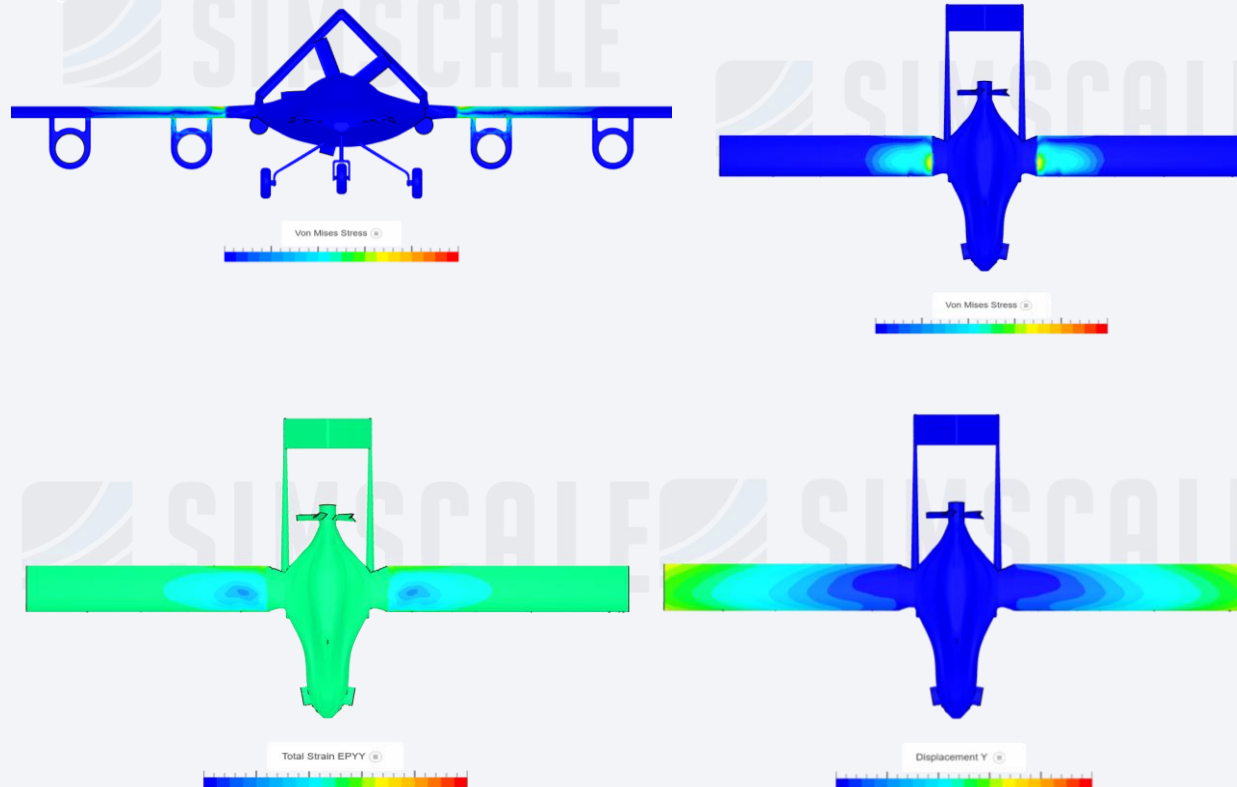
TECHNICAL SPECIFICATIONS

CAD Platform	Siemens NX
Fuselage Frame	Parametric Design
Body Method	3D Form Sub-D surface modelling
Surface Quality	High-quality Sub-D surfaces with G1 curvature
Assembly	Full system integration
Rendering	NX Render + post-processing

Static Structural Analysis: UAV

PROJECT OVERVIEW

Static analysis on the fixed-wing UAV was carried out using Simscales, whereby 981 N loads were applied vertically downwards to each of the four underwing missile pylons giving a total of 3,924 N forces (400kg payload with each missile weighing , with the wings being restrained at their roots. The results from von Mises stresses show that there is no stress concentration outside the wing root intersections and there is no material yielding within the airframe. The results obtained from displacement magnitudes show that the structure behaves according to cantilever bending theory with a consistent symmetric gradient of displacements throughout the length of both



KEY FEATURES & HIGHLIGHTS

- 1 Fixed supports applied at both wing roots with 981 N downward point loads at each of the four missile pylons, totaling 3,924 N.
- 2 Von Mises stress confirms no yielding across the airframe with peak stress concentrations symmetrically localised at both wing root junctions under 3,924 N total missile payload loading.
- 3 Displacement magnitude and Displacement Y confirm classic cantilever bending with maximum deflection at the wingtips and near-zero displacement at the fixed wing roots.
- 4 Total Strain EPXX reveals peak axial strain at the wing root pylon attachments, consistent with bending-induced tension and compression either side of the neutral axis.
- 5 Displacement Z and Total Strain EPYY confirm transverse and spanwise strain concentrations at the inboard pylon locations, validating the wing root junction as the critical structural region.

04

SEDAN

Surface Modelling · FEA · CFD Analysis

SEDAN CAR

PROJECT OVERVIEW

A sedan car was designed in Siemens NX, then subjected to independent FEA and CFD simulation studies in Ansys 2024 R2.



KEY FEATURES & HIGHLIGHTS

- 1 Surfaces of exterior body parts designed on Siemens NX through parametric and Class-A surface design approach
- 2 Curvature G2 continuity is maintained throughout panel transitions confirmed by zebra pattern evaluation
- 3 Structural body parts such as body Monocoque, subframe, and suspension attachments designed by simplified parametric curves

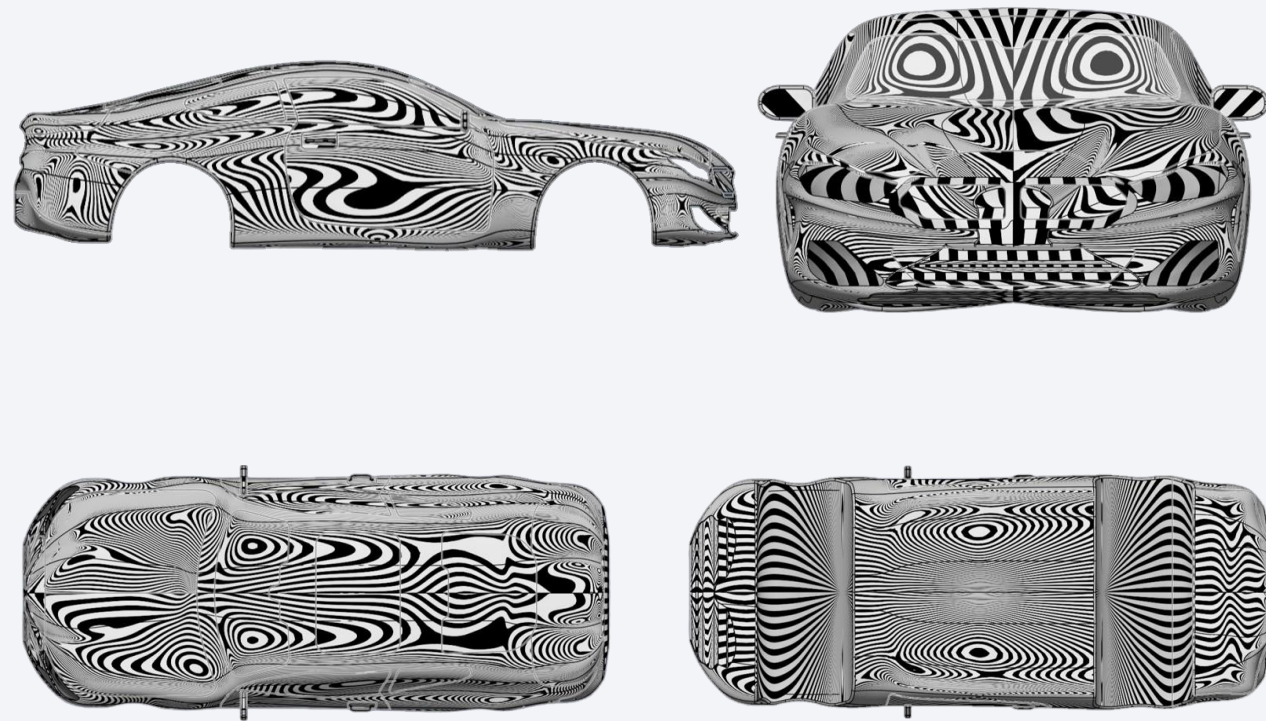
TECHNICAL SPECIFICATIONS

CAD Platform	Siemens NX
Monocoque	Parametric Design
Body Method	3D Form Sub-D surface modelling
Surface Quality	High-quality Sub-D surfaces with G1 and G2 curvature
Assembly	Full system integration
Rendering	NX Render + post-processing

SEDAN CAR

PROJECT OVERVIEW

A sedan car was designed in both SOLIDWORKS and Siemens NX, then subjected to independent FEA and CFD simulation studies in Ansys 2024 R2.



KEY FINDINGS

Zebra stripe analysis confirms G2 curvature continuity across the roofline, bonnet flanks, sill geometry, and underfloor panel, with smooth and organised stripe flow observed across all four orthographic views.

1

Concentric curvature patterns at the bonnet c and roof indicate uncontrolled compound curvature at the loft surface, identified as an area for iterative refinement through additional guide curve control.

2

The underfloor surface demonstrates the strongest curvature continuity across the entire model, with clean stripe flow maintained across the full floor panel and transmission tunnel geometry.

3

Panel boundary transitions achieve G1 tangency continuity at shut lines and feature edges, consistent with production automotive surfacing practice where sharp geometric definitions are intentional rather than surface quality failures.

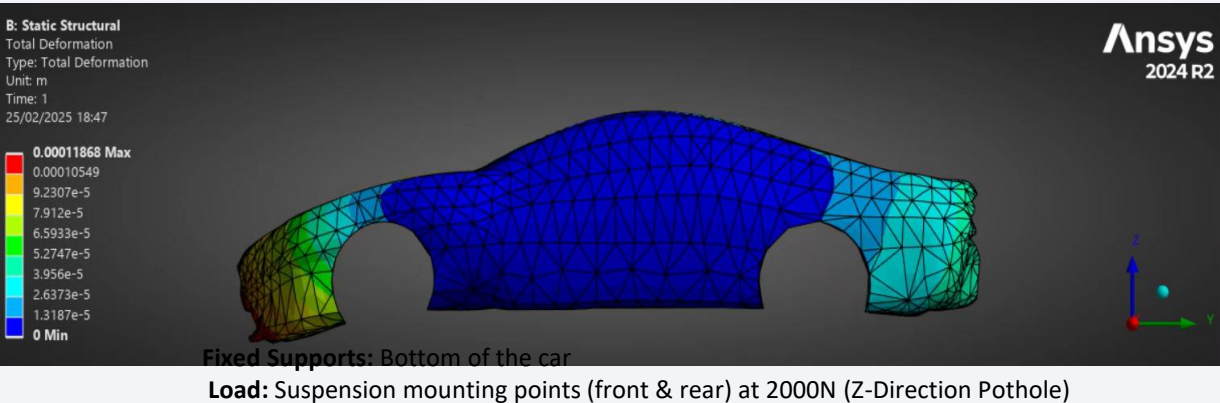
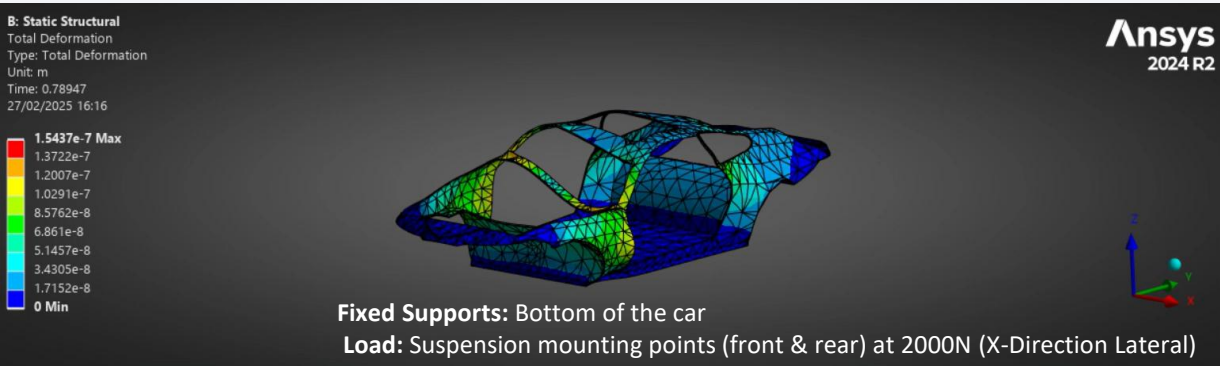
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Stress Analysis: Sedan Car

Body-in-White: Structural Steel
•Young's Modulus: 210 GPa
•Poisson's Ratio: 0.3

PROJECT OVERVIEW

A sedan car was designed in both SOLIDWORKS and Siemens NX, then subjected to independent FEA and CFD simulation studies in Ansys 2024 R2.



KEY FEATURES & HIGHLIGHTS

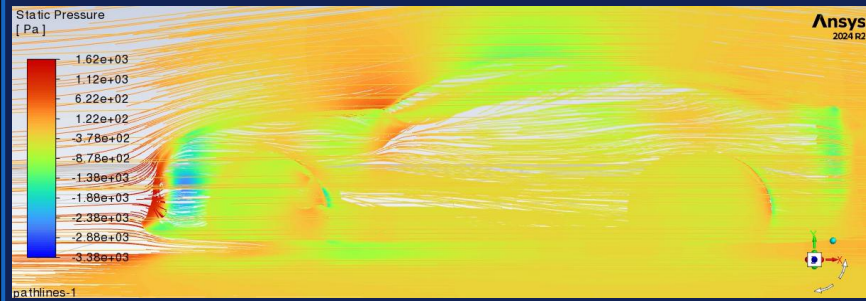
- 1 FEA: Max total deformation 0.11868 mm at highway structural load case (body shell)
- 2 FEA: Body-in-white mesh with tetrahedral elements and inflation at stress risers
- 3 CFD: Stagnation pressure 1620 Pa at front; negative pressure 3380 Pa at rear
- 4 CFD: Turbulent intensity highlights wake separation and underbody diffuser performance
- 5 CFD: X-velocity contours show boundary-layer development along roofline and boot

TECHNICAL SPECIFICATIONS

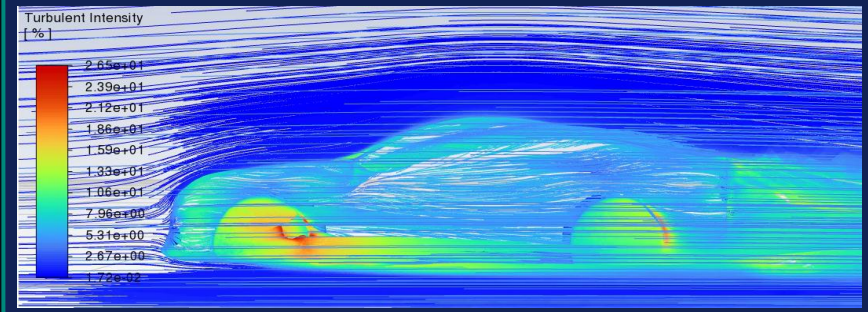
FEA Solver	Ansys Static Structural 2024 R2
FEA Output	Total deformation [m]
CFD Solver	Ansys Fluent 2024 R2
CFD Outputs	Pressure · X-Vel · Turbulence
Mesh Type	Tetrahedral mesh with local refinement and inflation layer
Flow Speed	51.5 m/s (X-velocity freestream)

CFD Simulation: Sedan Car

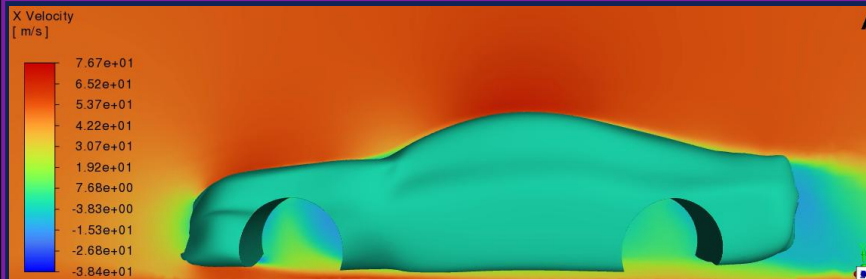
Static Pressure



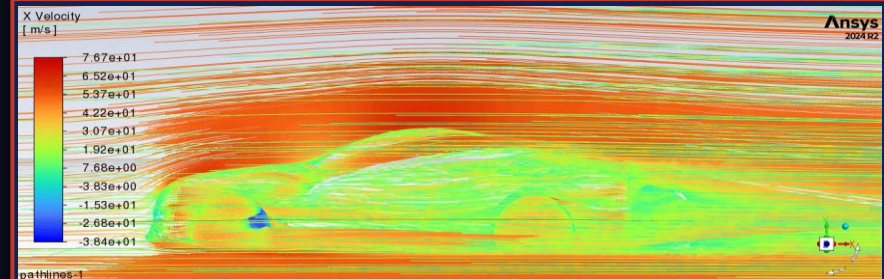
Turbulence Intensity



X-Velocity



X-Velocity



THANK YOU

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